

UNIT 1: Part 1

Foundations of Computer Networks: Evolution of computer networks, basic principles of network design, network topologies (Star, Mesh, Bus, Ring, Hybrid)

Introduction to Computer Networks

- A **computer network** is a system in which two or more autonomous computing devices are interconnected through communication channels to exchange data, share resources, and coordinate activities.
- These devices may include computers, servers, smartphones, printers, routers, switches, and IoT devices.
- Communication in a computer network is governed by **protocols**, which define rules for data formatting, transmission, reception, and error handling.
- Networks can be classified based on size (LAN, MAN, WAN), architecture (client-server, peer-to-peer), and transmission medium (wired or wireless).
- Computer networks enable global connectivity and form the foundation of modern technologies such as the **Internet, cloud computing, e-commerce, online education, telemedicine, and social networking**.

Need for Computer Networks

The growing demand for information sharing and communication has made computer networks indispensable. The major needs are explained below:

1. Resource Sharing

Computer networks allow multiple users to share:

- **Hardware resources** (printers, scanners, storage devices)
- **Software resources** (licensed applications, databases)
- **Data and files**

This eliminates redundancy and improves efficiency.

2. Communication and Collaboration

Networks provide fast and reliable communication through:

- Email and instant messaging
- Audio/video conferencing
- Collaborative platforms (shared documents, remote teamwork)

This is essential for businesses, education, and research.

3. Cost Reduction

By sharing resources:

- Organizations reduce hardware investment
- Maintenance and licensing costs are minimized
- Centralized servers lower operational expenses

4. Reliability and Fault Tolerance

Networks improve reliability by:

- Providing **backup servers**
- Using **redundant paths** for data transmission
- Ensuring data availability even if one component fails

5. Scalability and Flexibility

Networks can easily grow by:

- Adding new nodes or users
- Expanding coverage without redesigning the entire system

This supports organizational growth.

6. Centralized Management and Security

Network administrators can:

- Enforce security policies
- Control user access
- Monitor performance and detect faults centrally

7. Remote Access

Users can access data and services:

- From different geographical locations
- Through VPNs, cloud platforms, and remote desktops

Components of a Computer Network

A computer network is composed of various hardware and software components that collectively enable communication, data exchange, and resource sharing between devices. Each component plays a specific and critical role in network operation.

1. End Devices (Hosts / Nodes)

End devices are the **origin and destination points** of data communication in a network.

Examples

- Desktop computers, Laptops, Servers, Smartphones, Printers
- IoT devices (sensors, smart cameras)

Functions

- Generate data requests and responses
- Run applications such as browsers, email clients, and databases
- Access shared resources like files and printers

Types

- **Client devices:** Request services (PCs, mobiles)
- **Server devices:** Provide services (web servers, file servers)

2. Network Interface Card (NIC)

A **Network Interface Card** is a hardware component that enables a device to connect to a network.

Types

- Wired NIC (Ethernet), Wireless NIC (Wi-Fi)

Functions

- Converts digital data into electrical, optical, or radio signals
- Assigns a **unique MAC address** to each device
- Controls data transmission and reception

Without a NIC, a device cannot participate in network communication.

3. Transmission Media

Transmission media provide the **pathway** through which data travels.

A. Guided Media (Wired)

i) Twisted Pair Cable

- Used in LANs
- Low cost
- Susceptible to interference

ii) Coaxial Cable

- Better shielding than twisted pair
- Used in cable TV and older networks

iii) Optical Fiber

- Uses light signals
- Very high speed
- Immune to electromagnetic interference

B. Unguided Media (Wireless)

- Radio waves (Wi-Fi, Bluetooth)
- Microwaves (satellite communication)
- Infrared (short-range communication)

4. Networking Devices

Networking devices control data traffic and connect multiple devices.

a) Hub

b) Switch

c) Router

d) Bridge

e) Repeater

f) Modem

5. Network Software

Network software manages and controls network operations.

Types

- Network Operating Systems (Windows Server, Linux)
- Device drivers
- Network utilities

Functions

- User authentication
- Resource sharing
- Network monitoring and management

6. Network Protocols

Protocols define **rules and standards** for communication.

Common Protocols

- TCP/IP – Internet communication
- HTTP/HTTPS – Web browsing
- FTP – File transfer
- SMTP/POP3 – Email communication

Functions

- Data formatting

- Error detection and correction
- Flow control and congestion control

7. Network Services

Services are applications delivered through the network.

Examples

- File and print services
- Email services
- Web hosting
- Cloud storage
- Database services

8. Network Topology

Topology refers to the physical or logical layout of the network.

Types

- Bus
- Star
- Ring
- Mesh
- Tree
- Hybrid

Importance

Topology affects performance, reliability, and scalability.

9. Security Components

Security components protect the network from threats.

Includes

- Firewalls
- Antivirus and anti-malware

- Intrusion Detection Systems (IDS)
- Encryption techniques

10. Addressing and Naming Mechanisms

Used to uniquely identify devices.

Types

- MAC Address – Physical identification
- IP Address – Logical identification
- DNS – Converts domain names to IP addresses

Evolution of Computer Networks

The evolution of computer networks reflects the continuous growth in computing power, communication technologies, and user requirements. From isolated mainframe systems to today's intelligent, high-speed, and globally connected networks, computer networks have undergone several important stages.

Timeline of Computer Network Evolution

Period	Stage	Key Developments
1950s–1960s	Pre-Network Era	Standalone computers, no networking
1969	ARPANET	First packet-switched network
1970s	Protocol Development	TCP/IP, email
1980s	LAN Era	Ethernet, Token Ring
1990s	Internet & WWW	Public Internet, web browsers
2000s	Wireless Networks	Wi-Fi, 2G–4G
2010s–Present	Modern Networks	Cloud, IoT, 5G, SDN

1. Pre-Network Era (1950s – Early 1960s)

Before networking, computers worked in isolation.

Characteristics

- Standalone computers
- Data shared using punched cards, magnetic tapes, or floppy disks
- No real-time communication
- High operational cost

Limitations

- Inefficient data sharing
- Duplication of work
- Poor resource utilization

This created the need for interconnected computing systems.

2. First Generation Networks – ARPANET (1960s)

The first significant step toward networking was the development of **ARPANET**.

Key Features

- Developed by the U.S. Department of Defense (DARPA)
- Used **packet switching** instead of circuit switching
- Connected universities and research institutions
- Designed for reliability during failures

Importance

- Foundation of the Internet
- Proved that packet-switched networks are feasible

3. Second Generation – Protocol Development (1970s)

This period focused on standard communication methods.

Key Developments

- Introduction of **TCP/IP**
- Separation of communication into layers
- Development of email as a major application
- Internetworking concept introduced

Impact

- Different networks could communicate
- Improved reliability and scalability

4. Third Generation – Local Area Networks (1980s)

Networking became common within organizations.

Key Technologies

- **Ethernet (IEEE 802.3)**

- Token Ring
- Fiber Distributed Data Interface (FDDI)

Characteristics

- High-speed data transmission
- Networking within offices, campuses, and industries
- File and printer sharing

Outcome

- Reduced computing costs
- Increased productivity

5. Fourth Generation – Internet and World Wide Web (1990s)

This era marked the public explosion of networking.

Key Innovations

- Global expansion of the Internet
- Introduction of the **World Wide Web (WWW)**
- Web browsers (e.g., Mosaic, Netscape)
- Growth of email, e-commerce, and online services

Impact

- Networking became accessible to the general public
- Businesses moved online
- Information became globally available

6. Fifth Generation – Wireless and Mobile Networks (2000s)

Mobility became a central focus.

Key Developments

- Wi-Fi (IEEE 802.11)

- Mobile networks: 2G → 3G → 4G
- Smartphones and portable devices

Features

- Wireless Internet access
- Anytime, anywhere connectivity
- Rapid growth of mobile applications

7. Modern Networks – Cloud, IoT, and 5G (2010s – Present)

Current networks emphasize speed, intelligence, and scalability.

Major Advancements

- **Cloud computing** (IaaS, PaaS, SaaS)
- **Internet of Things (IoT)** – billions of connected devices
- **5G networks** – high speed and low latency
- **Software-Defined Networking (SDN)**
- Network virtualization

Benefits

- On-demand resources
- Massive device connectivity
- Support for AI, big data, and real-time applications

8. Future Trends in Computer Networks

Expected Developments

- 6G networks
- AI-driven network management
- Quantum networking
- Ultra-low latency communication

- Enhanced security architectures

Basic Principles of Network Design

Network design refers to the planning and implementation of a computer network in such a way that it meets current requirements while remaining efficient, secure, reliable, and scalable for future needs. A well-designed network ensures optimal performance, ease of management, and cost-effectiveness.

1. Scalability

Scalability is the ability of a network to **grow and adapt** to increasing numbers of users, devices, and applications without major changes to the existing infrastructure.

Key Points

- Supports future expansion
- Avoids complete network redesign
- Ensures consistent performance

Methods to Achieve Scalability

- Hierarchical network design (core, distribution, access layers)
- Use of modular components
- Adoption of standard protocols

2. Reliability and Availability

Reliability ensures continuous network operation, while availability refers to the percentage of time the network remains functional.

Key Features

- Redundant links and devices
- Backup servers and power supplies
- Failover mechanisms

Importance

- Minimizes downtime
- Ensures uninterrupted services
- Critical for banking, healthcare, and enterprise networks

3. Performance

Performance refers to how efficiently the network transfers data.

Performance Metrics

- Bandwidth
- Throughput
- Latency
- Packet loss

Design Considerations

- Proper selection of network topology
- Load balancing
- Efficient routing and switching

4. Security

Security protects network data and resources from unauthorized access, attacks, and misuse.

Security Measures

- Authentication and authorization
- Encryption techniques
- Firewalls and intrusion detection systems
- Regular updates and patches

Objective

- Ensure confidentiality, integrity, and availability (CIA triad)

5. Manageability

Manageability refers to the ease of monitoring, configuring, and maintaining the network.

Key Aspects

- Centralized management tools
- Network monitoring and diagnostics
- Configuration management

Benefits

- Faster troubleshooting
- Reduced maintenance effort
- Improved network efficiency

6. Cost Effectiveness

Cost effectiveness ensures optimal network performance within budget constraints.

Considerations

- Hardware and software costs
- Maintenance and operational expenses
- Scalability versus initial investment

Goal

- Maximum performance at minimum cost

7. Standardization and Interoperability

Definition

Standardization ensures compatibility between devices from different manufacturers.

Examples

- TCP/IP protocol suite

- IEEE networking standards
- OSI model

Advantages

- Vendor independence
- Easy integration
- Long-term sustainability

8. Flexibility and Adaptability

Flexibility allows the network to adapt to new technologies and changing requirements.

Examples

- Support for wired and wireless integration
- Cloud and virtualization support
- Software-defined networking (SDN)

9. Quality of Service (QoS)

QoS ensures that critical applications receive priority in data transmission.

Applications

- Voice over IP (VoIP)
- Video conferencing
- Real-time data transfer

Techniques

- Traffic prioritization
- Bandwidth reservation
- Traffic shaping

10. Fault Tolerance

Fault tolerance is the ability of a network to continue functioning even when components fail.

Techniques

- Redundant hardware
- Multiple communication paths
- Automatic recovery mechanisms

11. Compliance and Policy Support

Network design must comply with organizational policies and legal regulations.

Examples

- Data protection laws
- Access control policies
- Audit and logging mechanisms

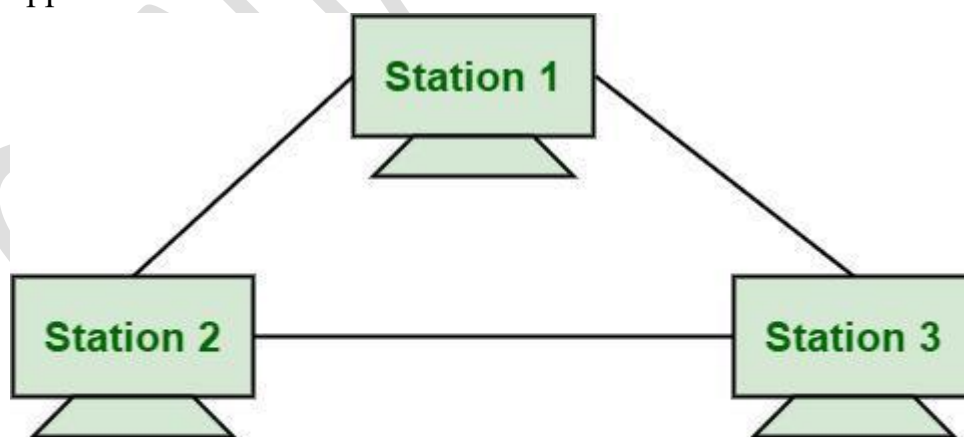
Classification of Computer Network on basis of Architecture

Computer networks can be classified based on architecture into two primary types:

- Peer-to-Peer Architecture
- Client/Server Architecture
- Hybrid Networks

1. Peer-to-Peer Architecture

- In a P2P network, computers (peers) are directly connected, usually via the Internet, to share files and resources without a central server.
- Each computer acts as both client and server, with equal roles and responsibilities.
- Tasks are distributed across all devices, making P2P suitable for small setups (up to about 10 computers).
- Since there's no strict client-server division, peers can both send and receive data directly.
- P2P networks are commonly used in business, education, and military applications.



Peer-to-Peer Architecture

Advantages:

- **Cost-effective:** No need for expensive dedicated servers.
- **Easy to set up:** Relatively simple to configure.
- **Highly resilient:** No single point of failure.
- **Scalable:** Can easily add more nodes.

Disadvantages:

- **Security risks:** Difficult to enforce security policies.
- **Lack of central management:** Difficult to manage and monitor.
- **Performance issues:** Performance can degrade as the network grows.
- **Copyright issues:** Facilitates illegal file sharing.

Examples:

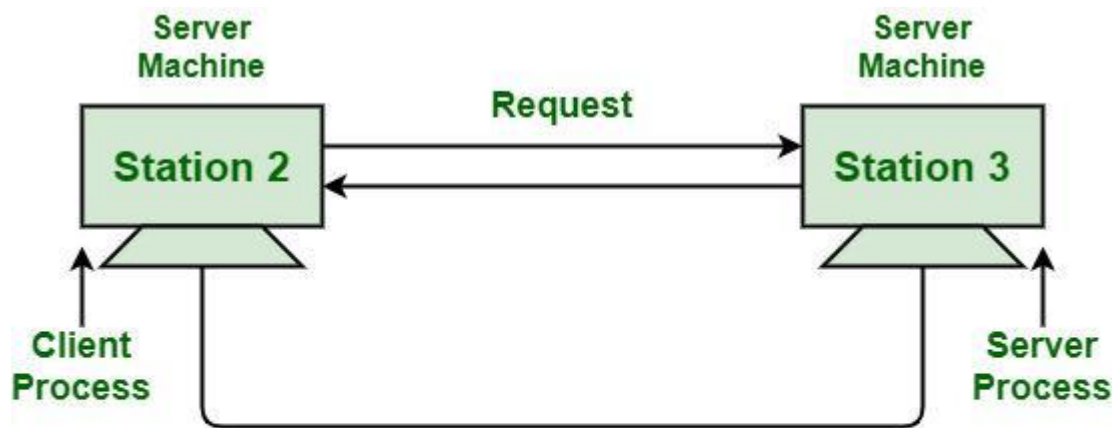
- File-sharing networks
- Cryptocurrency networks (e.g., Bitcoin)
- Collaboration tools

2. Client-Server Architecture

In a client-server architecture, there is a central server that provides resources and services to multiple clients. Clients request services from the server, and the server responds to those requests.

Characteristics:

- **Centralized:** A central server manages the network.
- **Dedicated Server:** A powerful computer dedicated to providing services.
- **Client Requests:** Clients send requests to the server.
- **Server Responses:** The server processes requests and sends responses to clients.
- **Scalability:** Can be scaled by adding more servers or upgrading existing ones.



Client/Server Architecture

Advantages:

- **Centralized management:** Easier to manage and monitor.
- **Improved security:** Easier to enforce security policies.
- **Better performance:** Dedicated servers can provide better performance.
- **Data backup and recovery:** Easier to back up and recover data.

Disadvantages:

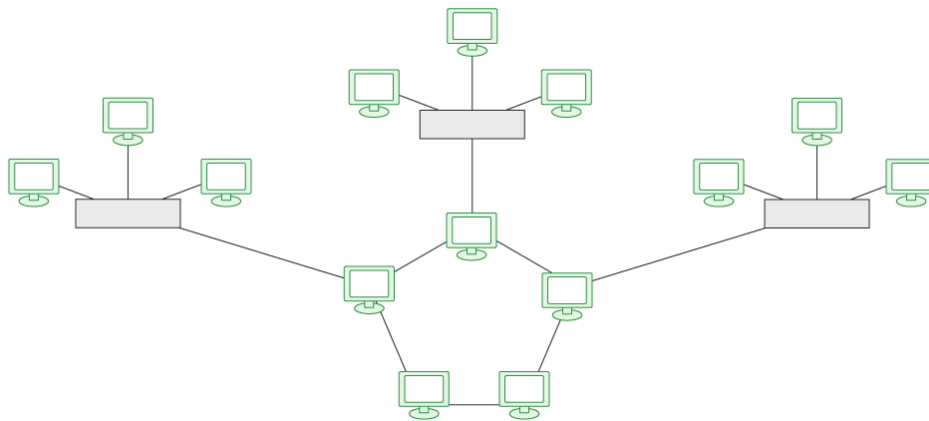
- **Costly:** Requires expensive dedicated servers.
- **Single point of failure:** Failure of the server can affect the entire network.
- **Scalability limitations:** Scaling can be complex and expensive.
- **Server overload:** The server can become overloaded if there are too many clients.

Examples:

- Web servers (e.g., Apache, Nginx)
- Email servers (e.g., Exchange, Sendmail)
- Database servers (e.g., MySQL, PostgreSQL)
- File servers

3. Hybrid Networks

- Hybrid networks combine elements of both client-server and peer-to-peer architectures.
- They leverage the centralized control of client-server networks and the decentralized resource sharing of P2P networks.
- An example is a torrent network, where a central tracker (server) coordinates peers, but file sharing occurs directly between peers.
- Hybrid networks are used in applications requiring both centralized management and distributed resource sharing.



Advantages of Hybrid Networks:

- Combines the benefits of centralized control and decentralized resource sharing.
- Flexible, allowing adaptation to different use cases and requirements.
- Can provide redundancy and fault tolerance by distributing tasks across peers and servers.

Disadvantages of Hybrid Networks:

- Complex to design and maintain due to the combination of two architectures.
- May face security challenges from both client-server and P2P components.
- Higher costs compared to pure P2P networks due to the need for server infrastructure.

Distributed Architecture

Distributed architecture involves distributing data and processing across multiple nodes or systems. This approach aims to improve performance, scalability, and fault tolerance. There are several types of distributed architectures:

1. Cloud Computing

Cloud computing involves delivering computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.

Characteristics:

- **On-demand self-service:** Users can provision resources as needed.
- **Broad network access:** Resources are accessible over the network.
- **Resource pooling:** Resources are shared among multiple users.
- **Rapid elasticity:** Resources can be scaled up or down quickly.
- **Measured service:** Usage is monitored and billed accordingly.

Advantages:

- **Cost savings:** Pay-as-you-go pricing model.
- **Scalability:** Easily scale resources up or down.
- **Flexibility:** Access a wide range of services.
- **Reliability:** High availability and fault tolerance.
- **Accessibility:** Access resources from anywhere with an internet connection.

Disadvantages:

- **Security concerns:** Data security and privacy risks.
- **Vendor lock-in:** Dependence on a specific cloud provider.
- **Latency issues:** Performance can be affected by network latency.
- **Compliance issues:** Meeting regulatory compliance requirements.

Examples:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

2. Microservices Architecture

- Microservices architecture is an approach to developing an application as a suite of small, independent services, built around a specific business capability.
- These services are independently deployable, scalable, and can be written in different programming languages.

Characteristics:

- **Independent deployability:** Each service can be deployed independently.
- **Scalability:** Each service can be scaled independently.
- **Technology diversity:** Different services can use different technologies.
- **Decentralized governance:** Each service can be managed independently.
- **Fault isolation:** Failure of one service does not affect other services.

Advantages:

- **Improved scalability:** Each service can be scaled independently.
- **Faster development:** Smaller codebases and independent deployments.
- **Technology flexibility:** Use the best technology for each service.
- **Fault isolation:** Failure of one service does not affect other services.
- **Easier maintenance:** Smaller codebases are easier to maintain.

Disadvantages:

- **Complexity:** More complex to design and manage.
- **Increased overhead:** More network communication and coordination.
- **Debugging challenges:** Debugging distributed systems can be challenging.

- **Data consistency:** Maintaining data consistency across services can be difficult.

Examples:

- Netflix
- Amazon
- Spotify

3. Edge Computing

- Edge computing involves *processing data closer to the source of the data*, rather than sending it to a central data centre.
- This reduces *latency and bandwidth usage*, and improves performance for applications that require real-time processing.

Characteristics:

- **Proximity:** Data processing occurs closer to the data source.
- **Low latency:** Reduced latency for real-time applications.
- **Bandwidth reduction:** Reduced bandwidth usage.
- **Autonomous operation:** Can operate independently of the central network.
- **Security:** Enhanced security by processing data locally.

Advantages:

- **Reduced latency:** Faster response times for real-time applications.
- **Bandwidth savings:** Reduced bandwidth usage.
- **Improved security:** Data is processed locally, reducing the risk of interception.
- **Autonomous operation:** Can operate even when the network is disconnected.
- **Scalability:** Can be scaled by adding more edge devices.

Disadvantages:

- **Complexity:** More complex to manage and deploy.

- **Security challenges:** Securing edge devices can be challenging.
- **Limited resources:** Edge devices have limited processing power and storage.
- **Management overhead:** Managing a large number of edge devices can be difficult.

Examples:

- IoT devices
- Autonomous vehicles
- Smart cities

Network Topology

Network Topology is the **structure or layout of a computer network** that defines how devices (nodes) such as computers, servers, switches, and routers are interconnected and how data flows between them.

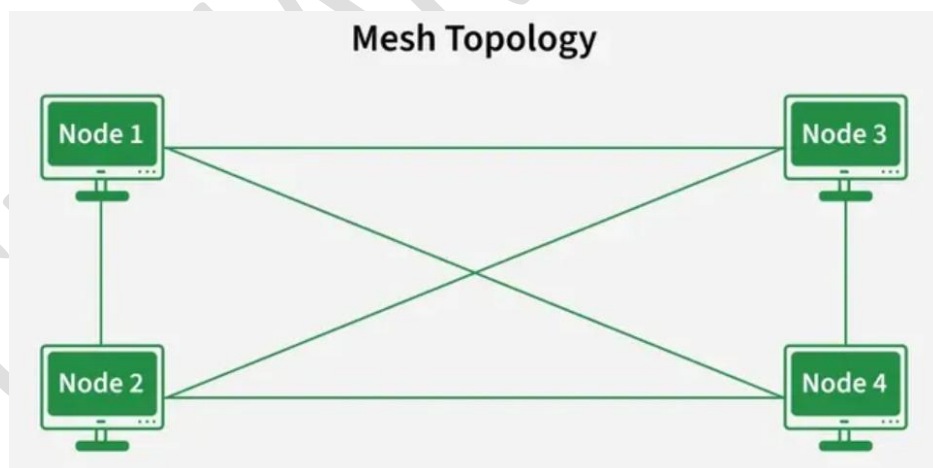
Topology can be **physical** (actual cable layout) or **logical** (path taken by data).

- **Physical Topology:** Physical topology indicates the arrangement of different elements of a network. It reflects the physical layout of devices and cables to form a connected network.
- **Logical Topology:** Logical topology defines the path that data packets take as they travel through the network independent of its physical layout.

Types of Network Topology

1. Mesh Topology

- In Mesh Topology, every node has a dedicated point-to-point link to every other node.



Mesh

- Suppose the **N** number of devices are connected with each other in a mesh topology:
 - The total number of ports that are required by each device is **N-1**.

- **The total number of ports required = $N * (N-1)$.**
- For example: There are 5 devices connected to each other, hence the total number of ports required by each device is 4 and the Total number of ports: 20
- Suppose, N number of devices are connected with each other in a mesh topology:
 - The **total number of dedicated links required to connect them i.e. $N(N-1)/2$.**
 - For example, In above figure, there are 5 devices connected to each other, hence the total number of links required is $5*4/2 = 10$.

There are two types of Mesh topologies:

- **Full Mesh Topology :**
 - **All the nodes within the network are connected with every other.**
 - If there are n number of nodes during a network, each node will have an n-1 number of connections.
- **Partial Mesh Topology :**
 - The partial mesh is more practical as compared to the full mesh.
 - **In a partially connected mesh, all the nodes aren't necessary to be connected with one another during a network.**

Advantages of Mesh Topology

- **Easy fault identification** and isolation.
- **Failure of a single device won't break the network.**
- There is no traffic problem, as there are dedicated point-to-point links for every computer.
- It provides **high privacy and security.**
- **Data transmission is more consistent** because failure doesn't disrupt its processes.
- Adding new devices won't disrupt data transmissions.

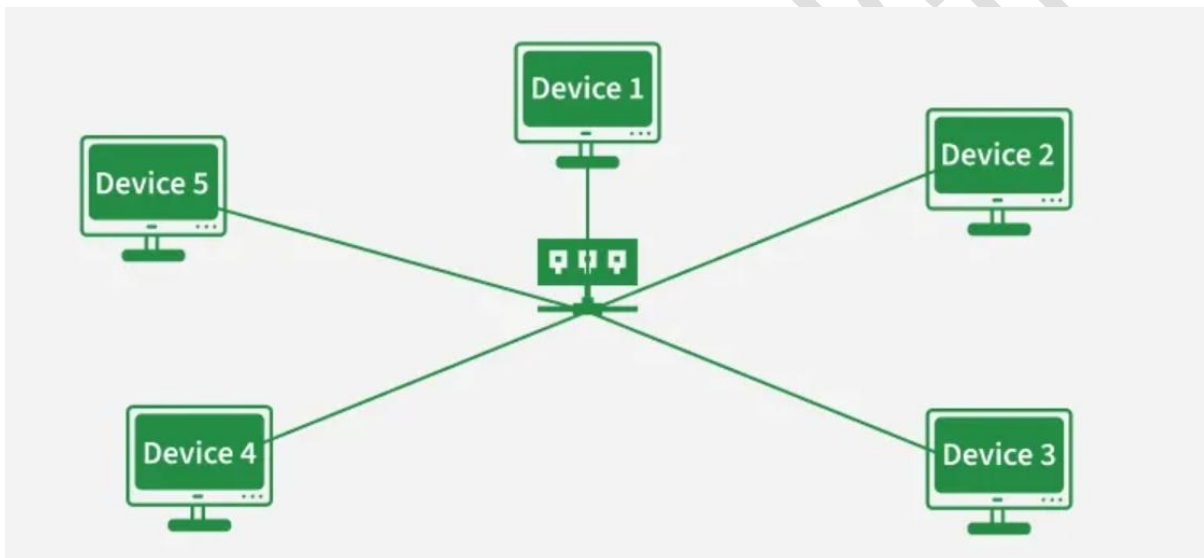
- A mesh doesn't have a centralized authority.

Disadvantages of Mesh Topology

- Expensive
- Complex installation
- High cabling overhead

2. Star Topology

- In a Star Topology, all the nodes (PCs, printers and peripherals) are **connected to the central server**.
- It has a central connection point, like a hub or switch. **In a star topology, each device is connected to the central hub.**



Advantages of Star Topology

- If N devices are connected to each other in a star topology, **then the number of cables required to connect them is N** . So, it is **easy to set up**.
- Each device requires only 1 port, i.e. to connect to the hub; therefore, the **total number of ports required is N** .
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

Disadvantages of Star Topology

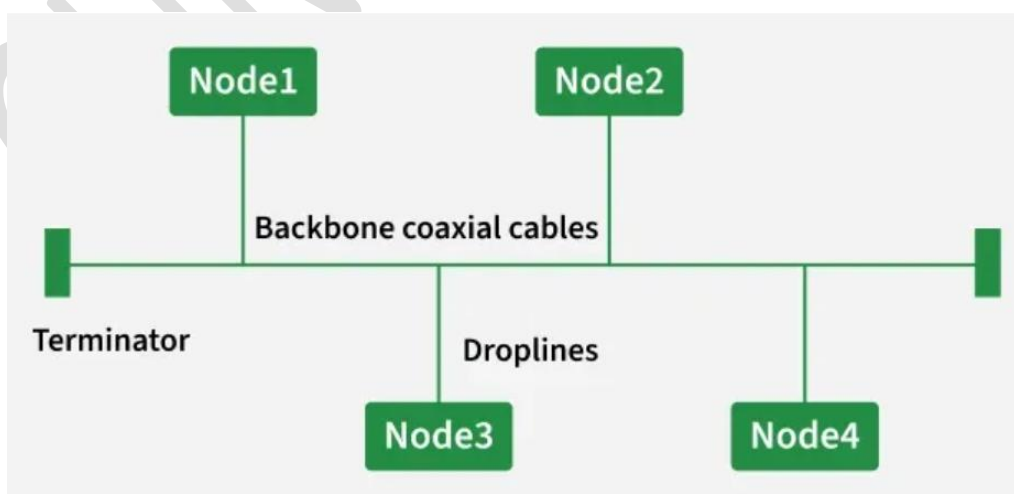
- Star networks **can require more cable length** than a linear topology.
- **More expensive** cabling.
- Performance is based on the single concentrator, i.e. hub.

Applications

- LANs
- Schools, offices, labs

3. Bus Topology

- In a **Bus topology**, all devices are connected to a **single main cable called the backbone**.
- Data sent by one device travels along the backbone and is received by all devices, but only the intended receiver accepts it.
- Thus, in the bus topology, signals are broadcast to all stations. Each computer checks the address on the signal (data frame) as it passes along the bus.
- If the signal's address matches that of the computer, the computer processes the signal. If the address doesn't match, the computer takes no action and travels down the bus.



Advantages of Bus Topology

- If N devices are connected to each other in a bus topology, then:
 - **The number of cables required to connect them is 1**, known as backbone cable,
 - **N drop lines** are required.
- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.
- **The cost of the cable is less compared to other topologies**, but it is used to build small networks.

Disadvantages of Bus Topology

- A bus topology is quite simpler, but still, it **requires a lot of cabling**.
- **If the common cable fails, then the whole system will crash down.**
- If the network traffic is heavy, it **increases collisions in the network**.
 - To avoid this, various protocols are used in the MAC layer known as **Pure Aloha, Slotted Aloha, CSMA/CD, etc.**
- Adding new devices to the network would slow down networks.
- Security is very low.

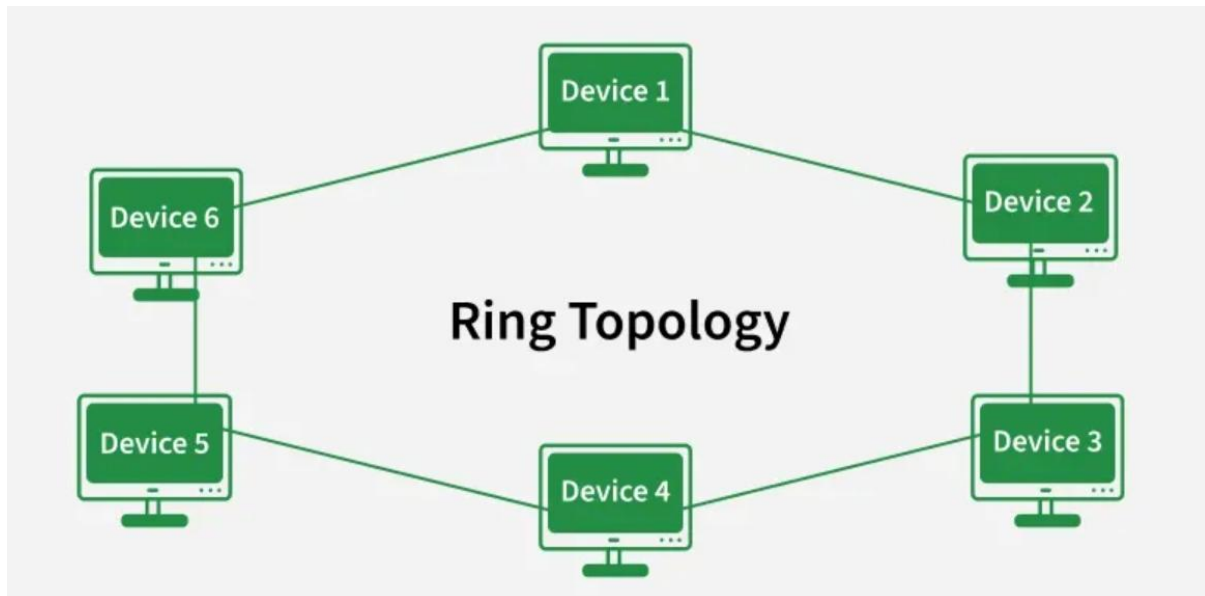
Applications

- Small, temporary networks
- Early Ethernet networks

4. Ring Topology

- All the nodes in a Ring Topology are **connected in a closed circle of cable**. Messages that are transmitted travel around the ring until they are addressed to, the signal being refreshed by each node.

- In a ring network, every device has exactly two neighbours for communication purposes.



The most common access method of ring topology is token passing.

- **Token passing:** It is a network access method in which a token is passed from one node to another node.
- **Token:** It is a frame that circulates around the network.

Advantages of Ring Topology

- The data transmission is **high-speed**.
- The **possibility of collision is minimum** in this type of topology.
- Cheap to install and expand.
- It is less costly than a star topology.

Disadvantages of Ring Topology

- The **failure of a single node in the network can cause the entire network to fail**.
- Troubleshooting is difficult in this topology.
- The **addition of stations in between or the removal of stations can disturb the whole topology**.

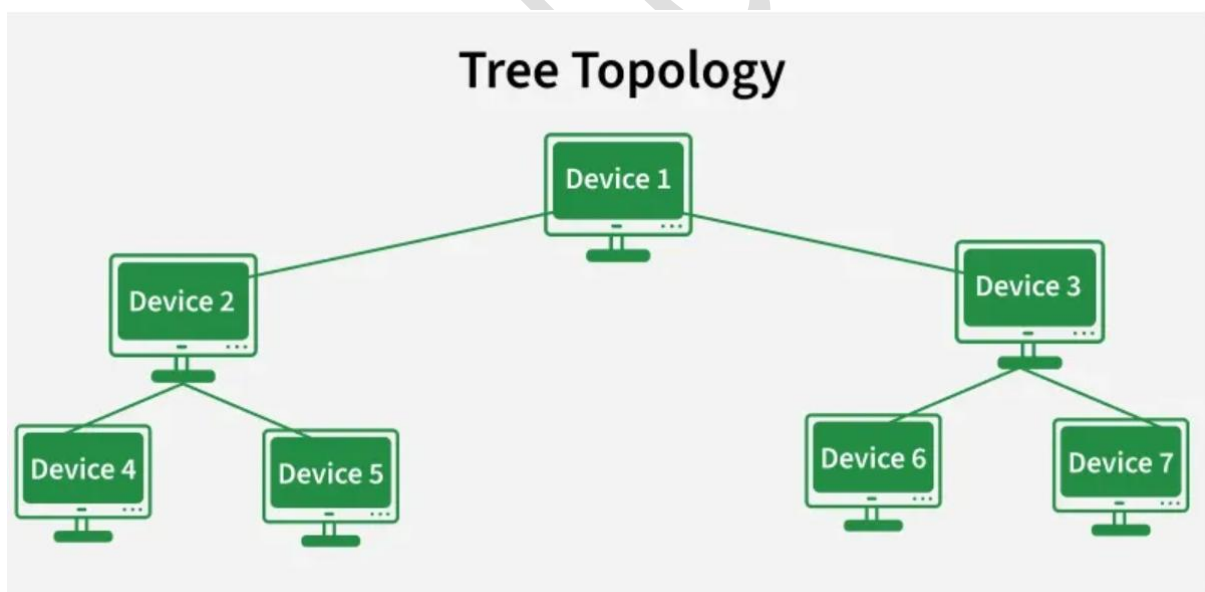
- Less secure.

Applications

- Token Ring networks
- FDDI

5. Tree Topology

- In tree topology nodes are connected in a **hierarchical structure to form a tree.**
- There is a root node in tree topology and the remaining nodes are considered as child nodes, basically it is a **combination of star and bus topology.**
- The central bus works as a communication pathway, and each star-configured network represents a level in the tree.
- In tree topology, a hierarchy is formed by the branching cable having no loops that connect the root with all other nodes for communication.



Advantages of Tree Topology

- **Security is high** in Tree Topology
- Tree Topology is **more reliable**
- Tree topology is **more scalable**

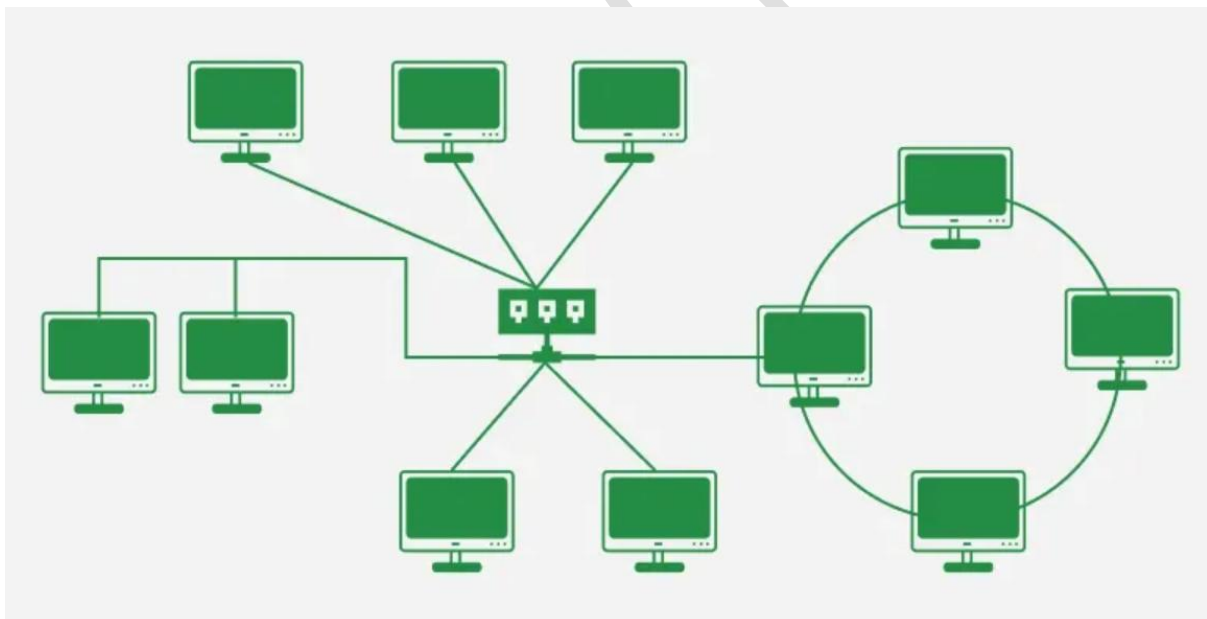
- It allows more devices to be attached to a single central hub thus it decreases the distance that is traveled by the signal to come to the devices.
- We can add new devices to the existing network.
- **Error detection and error correction** are very easy in a tree topology.

Disadvantages of Tree Topology

- If the central hub gets fails the entire system fails.
- **The cost is high** because of the cabling.
- **If new devices are added, it becomes difficult to reconfigure.**

6. Hybrid Topology

Hybrid topology is the combination of two or more types of topology, they arise from the integration of multiple network topologies that is why called Hybrid Network Topology.



Advantages of Hybrid Topology

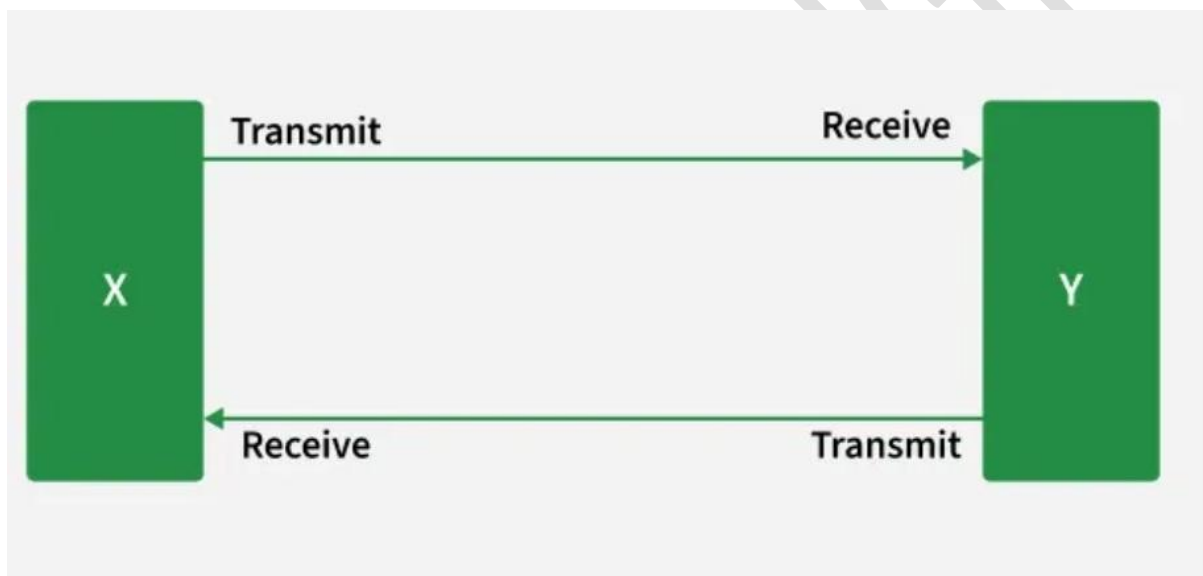
- Hybrid Topology provides more flexibility than others topologies.
- Hybrid topology is **more robust** than the other.
- Hybrid Topology provides **optimized performance**

Disadvantage of Hybrid Topology

- It is challenging to design the architecture of the Hybrid Network.
- **Hubs** used in this topology are very expensive.
- The infrastructure cost is very high as a hybrid network requires a lot of cabling and network devices.

7. Point to Point Topology

- Point-to-Point Topology is a type of topology that **works on the functionality of the sender and receiver.**
- It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver.
- Point-to-Point provides high bandwidth.



Advantages of Point to Point Topology

- **P2P networks are highly efficient as they allow for direct communication** between two devices without any intermediate devices or network components.
- **P2P networks are relatively more secure** than other topologies as they do not rely on intermediate devices that can be compromised or attacked.
- P2P networks are easy to configure and require minimal management or administration.

Disadvantages of Point to Point Topology

- **P2P networks are not scalable** as adding new devices requires establishing a separate link between each new device and the existing network, which can be time-consuming and expensive.
- **P2P networks can be difficult to maintain** as each device has to be managed separately.

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